Safety systems · Switzerland

SBB CFF FFS Rhätische Bahn

Swiss main signals (system L) show a **Fahrbegriff**, a speed for the route: FB 1 = line speed, FB 2 = 40, FB 3 = 60, FB 5 = 90, FB 6 = 40 "Kurze Fahrt" (short distance to a Halt). **On the RhB** the same aspects mean FB 2 = 30, FB 3 = 45, FB 6 = 30, and there is no FB 5. Distant signals announce the next main signal; Luzern – Sursee also has **system N** signals with numbers. Speeds in km/h. Quadrant controls are from TheJAG's profiles: RhB Allegra (Berninalinie), RhB Ge 4/4 II (Arosalinie), SBB RABe 523 (Luzern – Sursee).

Safety systems

Integra-Signum

SBB intermittent train protection (TSW: ETCS STM level)

A distant at Warnung (and a main signal at Kurze Fahrt) gives a buzzer and a yellow lamp: **acknowledge** at once (real: within 100 m or 5 s) or the train brakes. Passing a Halt brakes at once.

WHERE Luzern – Sursee (RABe 523)

QUADRANT **Gear lever** = Signum reset

ETCS Level 1

TSW: Staff Responsible, then Full Supervision (the real line runs Limited Supervision)

Select the level on the right display at a stand and confirm. Level 1 starts in SR (40 km/h to the first balise), then FS: speed monitoring and a braking curve with a warning tone. Level 0 only checks above 100 km/h.

WHERE Luzern – Sursee (RABe 523)

QUADRANT **ENG 1** = confirm the level; left aux = ETCS on

ZSI-90 / ZS

RhB Zugsicherung (TSW); today ZSI 127

Track magnets at a distant at Warnung and at Kurze Fahrt: two beeps and a flashing yellow lamp, press **ZS** or the emergency brake applies. Green lamp after a proceed aspect. Real RhB now runs balise-based ZSI 127.

WHERE Berninalinie, Arosalinie

QUADRANT Allegra: **ENG 1** = ZS, left aux = ZSI on. Ge 4/4 II: **gear lever** = ZSI acknowledge (always on)

SiSte / deadman

Sicherheitssteuerung, Totmannpedal

Allegra: SiSte, a timed vigilance like the SiFa: reset when the button lights. Ge 4/4 II: a deadman pedal, no timed alarm. RABe 523: none simulated.

WHERE Allegra, Ge 4/4 II

QUADRANT Allegra: **gear lever** = SiSte reset, right aux = SiSte on. Ge 4/4 II: left aux = deadman cut-out (lit = cut out)

Main signals (system L)

FDV five-lamp column: green, red, orange, green, green; FB 6 lights the second and fourth positions orange. RhB speeds in brackets.



Halt

Stop.

DRIVE Stop before the signal.



FB 1

Line speed.

DRIVE Drive on.



FB 2

Max 40 (RhB 30).

DRIVE Be at the speed at the signal.



FB 3

Max 60 (RhB 45).

DRIVE Be at the speed at the signal.



FB 5

Max 90 (not on the RhB).

DRIVE Be at 90 at the signal.

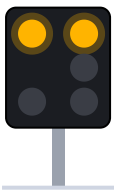


FB 6 Kurze Fahrt

Max 40 (RhB 30); a Halt follows at a short distance.

DRIVE Acknowledge Signum / ZS; be ready to stop.

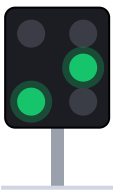
Distant signals (system L)



Warnung

The next main signal shows Halt.

DRIVE Acknowledge Signum / ZS; brake to stop.



Ankündigung FB 1

The next main signal shows line speed.

DRIVE Keep going.



Ankündigung FB 2 / 6

Next main signal: 40 (RhB 30).

DRIVE Brake to the speed.



Ankündigung FB 3

Next main signal: 60 (RhB 45).

DRIVE Brake to the speed.

Sources: Rivet Games: Berninalinie, Arosalinie and Luzern – Sursee driver's manuals · Schweizerische Fahrdienstvorschriften R 300.2 · RhB Betriebsvorschriften · de.wikipedia: Eisenbahnsignale in der Schweiz, Integra-Signum, Zugsicherung auf Schmalspurbahnen in der Schweiz

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System N (Luzern – Sursee)

One lamp; a number gives the speed in tens of km/h. Main signals have a white ring, distant signals a white square frame.



Green + 7

70 km/h from this signal.

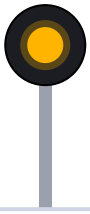
DRIVE Be at 70 here.



Orange + 7

70 km/h at the next signal.

DRIVE Brake to 70.



Orange

The next signal shows Halt.

DRIVE Acknowledge Signum; brake to stop.



Flashing number

Kurze Fahrt: a Halt follows at a short distance.

DRIVE Acknowledge; be ready to stop.

Speed signs and boards



Vorsignal verm. Geschw.

A lower speed ahead, in full km/h (two numbers: per train series).

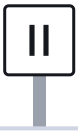
DRIVE Brake to reach it by the Anfangssignal.



Anfangssignal

The reduced speed starts here.

DRIVE Be at the speed here.



Endsignal

The reduced speed ends.

DRIVE Accelerate once the last vehicle has passed.



Merktafel Zugsicherung

At a distant: the train protection also acts at FB 2 / 3 here.

DRIVE Be ready to acknowledge.



Fehlendes Vorsignal

An entry signal follows at braking distance with no distant signal.

DRIVE Expect Halt.

How to drive

RABe 523: Signum and ETCS

1. At a stand select the ETCS level on the right display, then confirm with **ENG 1**.
2. Level 1: 40 km/h in SR to the first balise, then follow the FS speed and curve.
3. STM / Signum: at a Warnung press **Signum reset** (gear lever) at once.

RhB: ZS / ZSI

1. Switch the ZSI on (Allegra left aux; always on in the Ge 4/4 II).
2. At a distant at Warnung or a Kurze Fahrt: two beeps, yellow lamp flashing: press **ZS** (Allegra ENG 1, Ge 4/4 II gear lever).
3. Missed: red lamp and emergency brake; stop, release.

Vigilance

1. Allegra: SiSte on (right aux); reset with the **gear lever** when its button lights.
2. Ge 4/4 II: keep the deadman cut in: left aux **dark**.
3. Missed: emergency brake; stop, reset, release.

Before departure: Allegra: ZSI and SiSte on (both load off; aux squares lit). Ge 4/4 II: deadman cut in (left aux dark). RABe 523: ETCS on, level selected and confirmed. Check the signal, close the doors.

Sources: Rivet Games: Berninalinie, Arosalinie and Luzern – Sursee driver's manuals · Schweizerische Fahrdienstvorschriften R 300.2 · RhB Betriebsvorschriften · de.wikipedia: Eisenbahnsignale in der Schweiz, Integra-Signum, Zugsicherung auf Schmalspurbahnen in der Schweiz

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